

$$f_{X \sim}(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} (\lambda x)^{\alpha-1}$$

$$\boxed{x > 0}$$

$$f_{Y \sim}(z-x) = \frac{\lambda^\beta}{\Gamma(\beta)} e^{-\lambda(z-x)} (\lambda(z-x))^{\beta-1}$$

$$\boxed{z-x > 0} \rightarrow x < z$$

$$f_{X+Y}(z) = \int_0^z \frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} (\lambda x)^{\alpha-1} \cdot \frac{\lambda^\beta}{\Gamma(\beta)} e^{-\lambda(z-x)} (\lambda(z-x))^{\beta-1} dx$$

↑
pdf
for $X+Y$

$$= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha) \Gamma(\beta)} e^{-\lambda z} \int_0^z x^{\alpha-1} (z-x)^{\beta-1} dx$$

$$0 < x < z$$

$$= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha) \Gamma(\beta)} e^{-\lambda z} \int_0^1 (yz)^{\alpha-1} (z-yz)^{\beta-1} \frac{z}{z} dy$$

$$0 < \frac{x}{z} < 1$$

$$\frac{y = \frac{x}{z}}$$

$$= \frac{\lambda^{\alpha+\beta} z^{\alpha+\beta-1}}{\Gamma(\alpha) \Gamma(\beta)} e^{-\lambda z} \int_0^1 y^{\alpha-1} (1-y)^{\beta-1} dy$$

$$0 < y < 1$$

$$x = yz$$

$$dx = z dy$$

$$= \frac{C}{\Gamma(\alpha) \Gamma(\beta)} \lambda (\lambda z)^{\alpha+\beta-1} e^{-\lambda z}$$

$$\int_0^{\infty} \frac{c}{\Gamma(\alpha)\Gamma(\beta)} \lambda (\lambda z)^{\alpha+\beta-1} e^{-\lambda z} = 1.$$

$$\int_0^{\infty} \frac{\lambda}{\Gamma(\alpha+\beta)} (\lambda z)^{\alpha+\beta-1} e^{-\lambda z} = 1.$$

→

$$f_{X+Y}(z) = \frac{\lambda}{\Gamma(\alpha+\beta)} (\lambda z)^{\alpha+\beta-1} e^{-\lambda z}$$

↓

$$\text{Gamma}(\alpha+\beta, \lambda)$$

$$\begin{aligned}
& \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} \cdot \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{x^2}{2\sigma_1^2}} e^{-\frac{(z-x)^2}{2\sigma_2^2}} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{x^2}{2\sigma_1^2} - \frac{(z-x)^2}{2\sigma_2^2}} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{(\sigma_1^2 + \sigma_2^2)x^2 - 2\sigma_1^2 xz + \sigma_1^2 z^2}{2\sigma_1^2\sigma_2^2}} dx \\
&= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z \cdot x + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z^2}{2\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}}} dx \\
&= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-\frac{(\chi - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z)^2 - \frac{\sigma_1^4}{(\sigma_1^2 + \sigma_2^2)^2} z^2 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z^2}{2\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}}} dx \\
&= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-\frac{\frac{\sigma_1^4}{(\sigma_1^2 + \sigma_2^2)^2} - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}} z^2} \int_{-\infty}^{\infty} e^{-\frac{(\chi - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z)^2}{2\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}}} dx \\
&= \frac{1}{2\pi\sigma_1\sigma_2} e^{+\left(\frac{\frac{\sigma_1^4}{(\sigma_1^2 + \sigma_2^2)^2} - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}}\right) z^2} \int_{-\infty}^{\infty} e^{-\frac{(\chi - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z)^2}{2\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}}} dx
\end{aligned}$$

$$= \frac{1}{\sqrt{2\pi} \sqrt{\sigma_1^2 + \sigma_2^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \sqrt{\frac{\sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2}}} e^{-\frac{(x - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z)^2}{2 \left(\frac{\sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2} \right)^2}} dx$$

$$e^{-\frac{1}{2(\sigma_1^2 + \sigma_2^2)} z^2}$$

$$= \frac{1}{\sqrt{2\pi} \sqrt{\sigma_1^2 + \sigma_2^2}} e^{-\frac{1}{2(\sigma_1^2 + \sigma_2^2)} z^2}$$

$$Z \sim N(0, \sigma_1^2 + \sigma_2^2)$$

$$X_1 \sim N(\mu_1, \sigma_1^2)$$

$$X_2 \sim N(\mu_2, \sigma_2^2)$$

$$X_1 - \mu_1 \sim N(0, \sigma_1^2)$$

$$X_2 - \mu_2 \sim N(0, \sigma_2^2)$$

$$(X_1 - \mu_1) + (X_2 - \mu_2) \sim N(0, \sigma_1^2 + \sigma_2^2)$$

$$\underline{X_1 + X_2} \sim \underline{N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)}$$